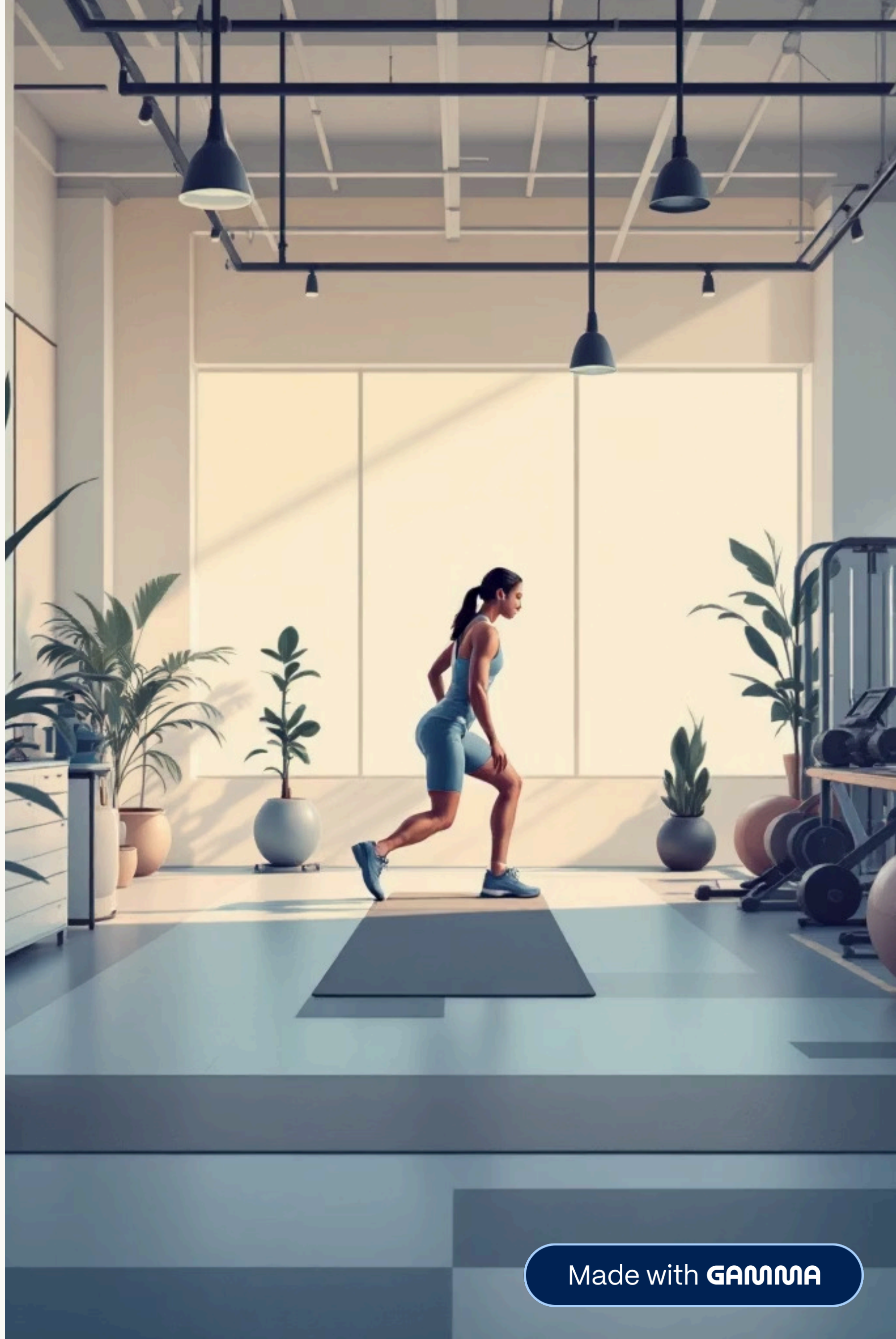
Fitness Intelligence Dashboard

Performance, Nutrition & Health Insights

Data-Driven Fitness & Health Analysis — **Ishan Shiwakoti**



Problem Statement

Why Fitness Data Needs Smarter Analysis

Understand Behavior

Analyze fitness, nutrition, and health data to decode user performance patterns and behavioral trends.

Identify Efficiency

Pinpoint the most effective workouts, dietary habits, and physiological health indicators driving results.

Detect & Improve

Uncover data inconsistencies and surface actionable opportunities to meaningfully improve fitness outcomes.

Dataset Overview

What the Data Contains

The dataset provides **user-level fitness and health records**, combining three critical dimensions into a single analytical view:

🏋️ Workout Performance

Exercise type, session duration, calories burned, sets & reps

🥗 Nutrition Metrics

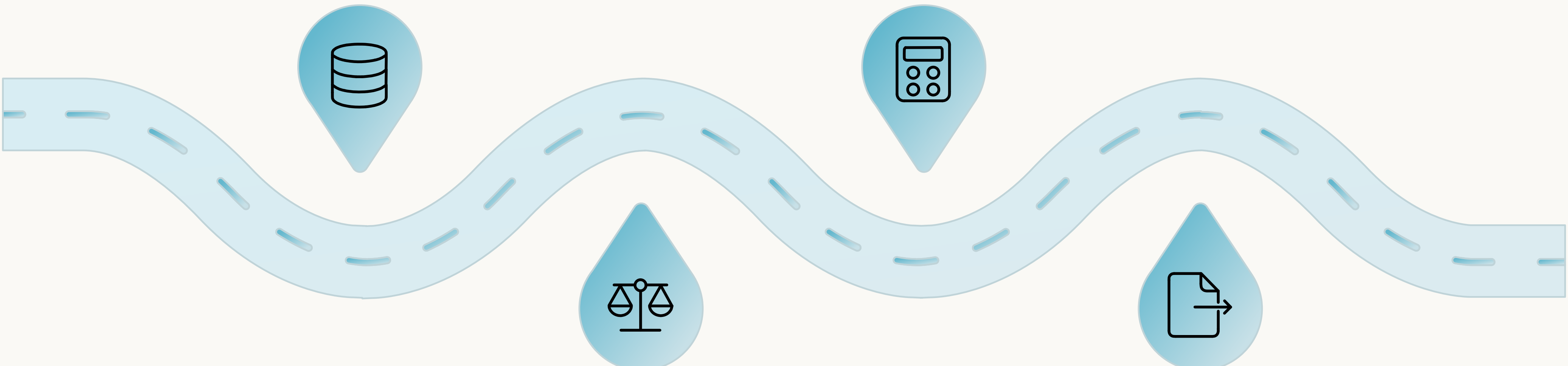
Meals, calorie intake, protein intake, serving size

❤️ Physiological Data

Heart rate (BPM), BMI, experience level, gender

Data Preparation

Cleaning & Structuring the Data



Missing Values

Identify and impute gaps

Standardize Units

Normalize calories, time, servings

Verify Calculations

Validate BMI and derived metrics

Prepare for Analysis

Finalize schema and export

Data quality was ensured by resolving missing entries, normalizing calories, duration, and serving sizes, and validating derived fields like BMI and calorie metrics — making the dataset fully ready for SQL querying and Power BI visualization.

Dashboard Overview

Interactive Fitness Intelligence in Power BI

Calories Burned

Session Duration

Avg BPM

BMI Aggregation

Nutrition Analysis

Key KPIs

Performance at a Glance

Calories Burned

Total calories burned across all workout sessions

Workout Duration

Cumulative session time across all users

Avg BPM

Mean heart rate reflecting workout intensity

Calorie Intake

Total dietary calorie consumption tracked

BMI Insights

Aggregate body mass index across user segments

Workout Performance

Which Workouts Deliver the Most Efficiency?

Not all workouts are created equal. Analysis reveals that **HIIT delivers the highest calorie burn per minute**, making it the most time-efficient option. Strength training leads in total session output.

Calories per Minute

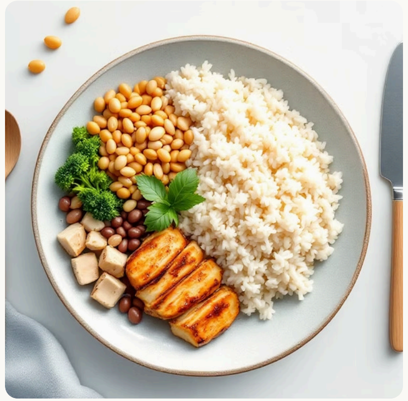
Core efficiency metric comparing workout types

Duration vs. Output

Longer sessions don't always mean more calories burned

Nutrition & Meal Analysis

Fueling Performance — or Hindering It?



Calorie Density

Identified meals with the **highest calories per gram** — key indicators for energy surplus risk.

Protein-Rich Meals

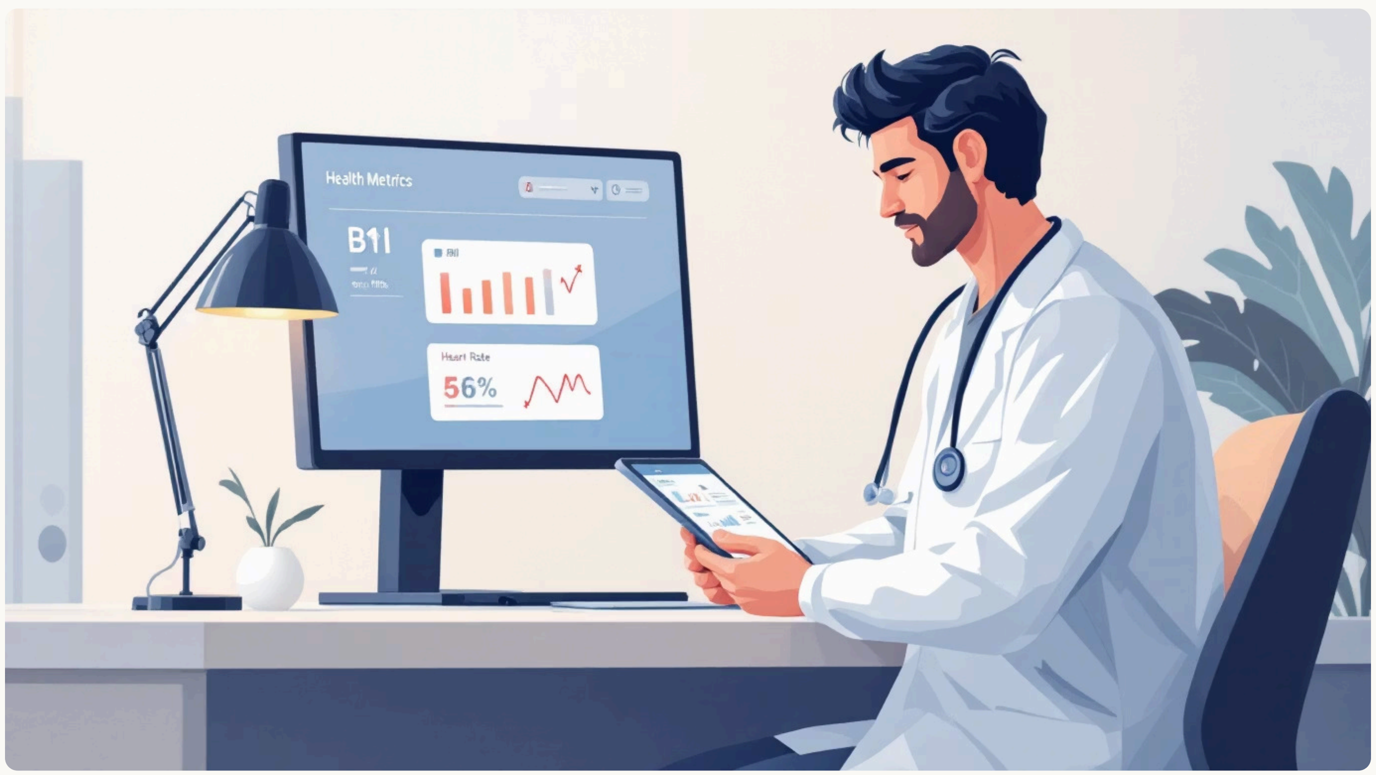
Flagged top **protein sources per serving** that support muscle recovery and fitness performance.

Dietary Patterns

Assessed how meal frequency and macro distribution **correlate with workout outcomes**.

Health Metrics

BMI & Heart Rate — The Full Picture



Physiological analysis surfaces meaningful disparities and flags potential data integrity concerns.

1

BPM by Gender

Compared average and peak heart rates across male and female users to assess cardiovascular intensity.

2

BMI Discrepancies

Detected gaps between **reported vs. calculated BMI** — signaling data quality or self-reporting issues.

Key Finding: BMI inconsistencies were found in a notable subset of users, warranting validation before clinical use.

Key Insights & Impact

What the Data Tells Us

🏆 Exercise Efficiency

Certain workouts deliver **significantly higher calorie burn per minute** — critical for optimizing training plans.

🍽️ Nutrition Risk

High calorie-density meals may **undermine fitness goals** when not aligned with energy expenditure.

📈 Experience Patterns

Advanced users exhibit **distinct workout and nutrition behaviors** — enabling targeted segmentation strategies.

✅ Balanced Outcomes

Users with **aligned nutrition and exercise habits** consistently demonstrate superior performance metrics.

These insights empower individuals to **optimize workout efficiency**, support smarter nutrition planning, and enable personalized, data-driven fitness strategies at scale.